

ODrive S1 Datasheet

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The ODrive S1 brings ODrive's next-generation motion control ecosystem to every robot, with a 2kW continuous power output and a wide variety of encoder interfaces, all at an accessible price point and with a form factor smaller than a credit card.

Electromechanical Specifications

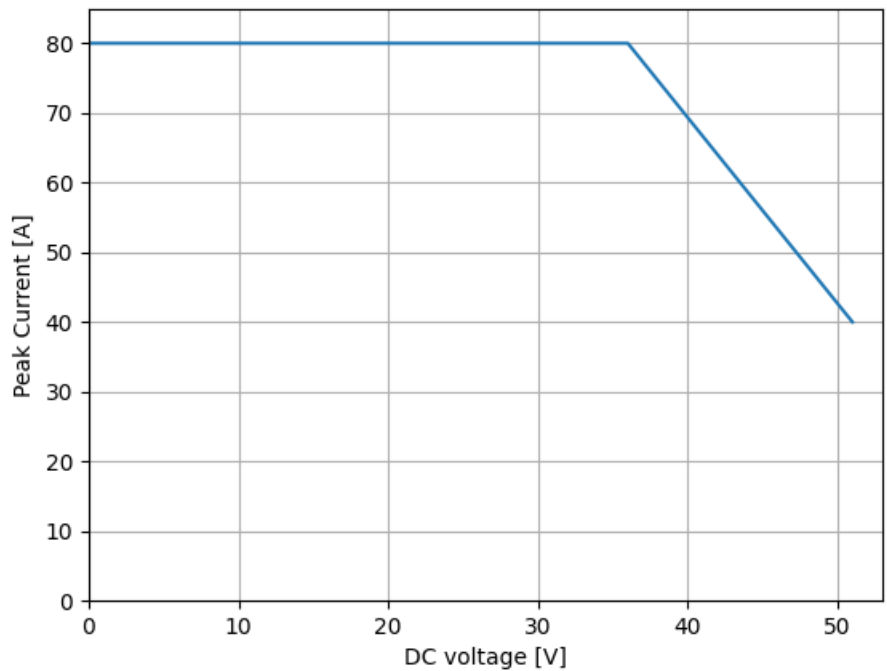
Electrical

Note

All specifications are in $0^{\circ}\text{C} \leq T_A \leq 40^{\circ}\text{C}$ unless otherwise noted.

Specification	Min.	Typ.	Max.	Units	Conditions and Notes
DC Voltage	12	16-48	50.5	V	
Maximum Modulation Depth		78%			Utilizable percentage of D
AUX Logic Voltage	10	12	14	V	Optional

Specification	Min.	Typ.	Max.	Units	Conditions and Notes
Operating Motor Current			20 40 40-80	A A A	Free air (T_A 25°C) Heat spreader plate (T_A 25°C) Peak, see below
ESD Protection		±30 ±13 ±8 ±8 ±6		kV	Power Lines, IEC 61000-4-2 CAN Lines, IEC 61000-4-2 Isolated Lines, IEC 61000-4-2 USB Lines, IEC 61000-4-2 All Other Lines, IEC 61000-4-2
CAN baudrate			8	Mbit/s	Additional firmware limits apply, see <code>baud_rate</code> .



ODrive S1 Peak Current vs DC voltage. The firmware limits motor current according to this curve depending on `dc_bus_overnoltage_trip_level`.

Connectors

Connector mating receptacles and crimps.

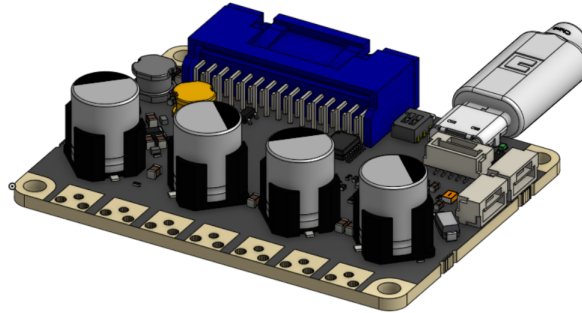
Connector	Description	Connector P/N	Mating Receptacle	Mating Crimp
Power	Optional Screw Terminal	TB005-762-07BE	N/A	N/A
J11	Unified I/O Header	S30B-PUDSS-1	PUDP-30V-S	SPUD-002T
J16, J17	CAN Header	SM04B-GHS-TB	GHR-04V-S	SSHL-002T
J1	Debug Header	BM05B-GHS-TB	GHR-05V-S	SSHL-002T

Environmental

Specification	Value	Notes
Humidity	Non-Condensing	
Ingress Protection	IP20	with case

CAD

The ODrive S1 CAD model is available on the [ODrive S1 OnShape](#) [↗] page.



Pinout

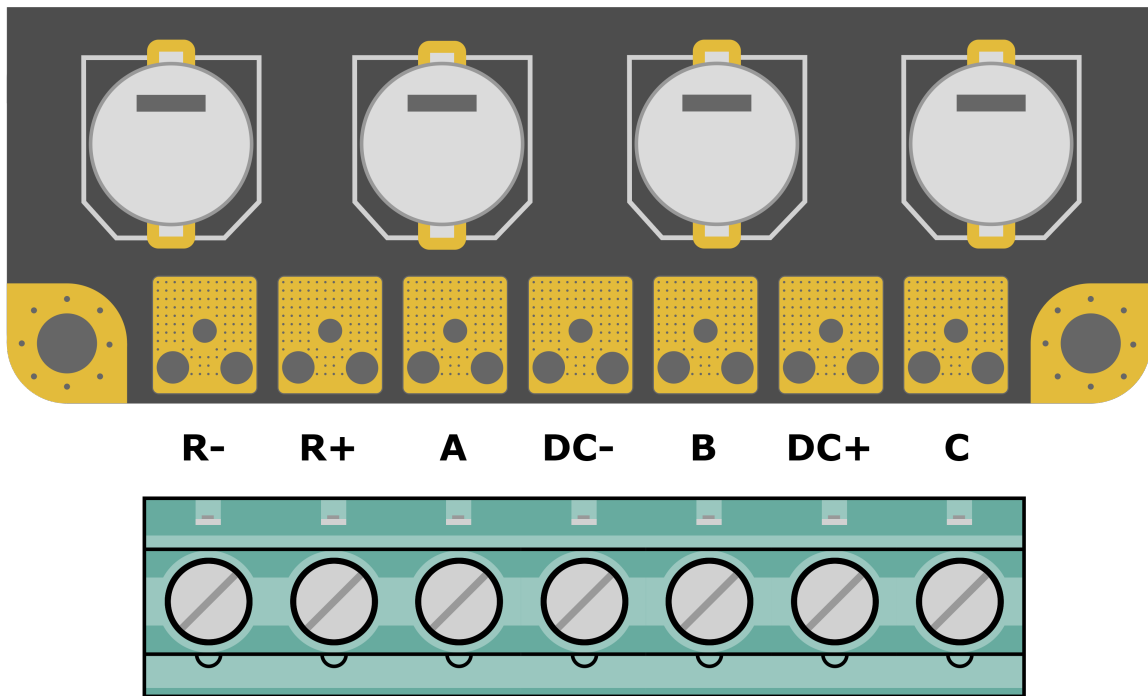
Power Pads

ⓘ Important

DC+- cannot tolerate reversed polarity, verify all power pad connections before energizing.

ⓘ Important

Do not connect multiple S1s to a single brake resistor! Each S1 must utilize its own separate brake resistor.



Pin	Description
R-	Brake resistor output
R+	Brake resistor output
A	Motor phase A
DC-	Power ground
B	Motor phase B
DC+	Power input, referenced to DC-
C	Motor phase C

The motor phase connections A/B/C can be connected in any order.

Logic Pins

- Digital mode is a general purpose mode that can be used for these functions: step, dir, enable, encoder index, hall effect encoder, SPI encoder nCS.
- All GPIO pins are 5V tolerant, except if you power the isolator with 3.3V, then **ISOLATED IO** inputs are rated to 3.3V nominal.

GPIO Config Legend

Fixed GPIO
"X"

Configurable
GPIO

Communication

Differential
Interface [4]

Feedback

Debug

Accessories

Analog Input

ODRIVE IO J11

ISOLATED IO

Requires external
power. See below
for more details.

CAN_H **
X
CAN_H **
X
12V IN [2]
X

ERROR
G06
UART TX
X
ISO VDD
G05
DIR

THERMISTOR +
G04
X
5V [1]
G03
ENC0 A
G02
HALL C
G01
HALL A
ANALOG IN
G0
+ RS 485 A
X
5V [1]
X
SPI MOSI
X
SPI SCK
X

1 3 5
2 4 6

7 9 11
8 10 12

13 15 17 19 21 23 25 27 29
14 16 18 20 22 24 26 28 30

X
CAN_L **
X
CAN_L **
X
3.3V [1]
X

G07
ENABLE
UART RX
X
ISO GND
G08
STEP
PWM IN

X
THERMISTOR -
GND
X
ENC0 B
G09
ENC0 Z
G10
HALL B
ANALOG IN
G11
RS485 B
G0
-
X
GND
X
SPI MISO
X
SPI nCS
G12

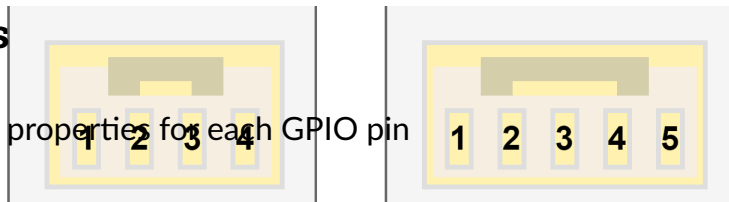
** Must
follow CAN
grounding
scheme.

CAN IO
J16, J17

DEBUG IO
J1

GPIO Properties

Additional notes and properties for each GPIO pin



❗ Note

The letter G and the zero padding are not used in `odrivetool` or the [web GUI](#), i.e. G09 would be represented by only the number 9.



- **G0:** This pin is driven at RS422 logic levels, and cannot be used as a logic input. This pin cannot be used if an external SPI encoder is used. The + and - pins are complementary outputs.
- **G01, G02, G11:** The HALL pins have a 2.7k Ω pullup to +5V and low pass filter ($\tau=4.25\mu s$), and cannot be used in DIGITAL_PULL_DOWN mode. When used as an output, the logic low level will be 500mV.
- **G03, G09, G10:** The ENCO pins have a 2.7k Ω pullup to +5V, and cannot be used in DIGITAL_PULL_DOWN mode. When used as an output, the logic low level will be 500mV.
- **G04:** The THERMISTOR+ pin has an internal 1k Ω pullup to 3.3V for use in a thermistor sense circuit. It cannot be used in DIGITAL_PULL_DOWN mode.
- **G12:** This pin can only be used as a SPI nCS pin, and should not be configured as a user-controlled input or output.
- **G05, G07, G08:** See [ISOLATED IO](#) section for required information using isolated pins. These pins are connected to a digital isolator, and can be used as GPIO inputs only. There is a 1.5M Ω pull-down integrated to the isolator, and pull up/down configuration has no effect.
- **G06:** See [ISOLATED IO](#) section for required information using isolated pins. This pins is connected to a digital isolator, and can be used as a GPIO output only.

User Facing Pins (Gxx)

- Locations for all pins that can be configured using `GpioMode`.

• Inputs:

G0-G03, G05, G07-G11

• Outputs:

G0-G03, G06, G09-G11

❗ Note

Inputs and outputs are not mutually exclusive.

Notes

- 5V outputs: combined draw max 150mA
- 3.3V output: max draw 150mA
- [1] 5V and 3.3V can be switched on and off internally. (feature coming soon!)
- The RS485 and SPI feedback interfaces are mutually exclusive on S1 – **a RS485 and external SPI encoder cannot be used at the same time**
- THERMISTOR + has an integrated 1k ohm [voltage divider](#). 
- [2] 12V in is optional and is used to power the ODrive logic, enabling communication before the main power supply is connected. Allowed voltage range: 10V-14V. Referenced to DC-. This voltage range is a hard limit, and inrush and overvoltage protection must be guaranteed.
- [3] GND NC offers a common connection for the CAN bus ground; it is not connected internally.
- [4] Differential interface used for RS-485 encoders.

ISOLATED IO (G05-G08)

- GND ISO and V+ ISO must be connected to your other board.
- V+ ISO is a power **Input** to the isolated interface, which you should power with 3.3V or 5V.
- Input and output levels are 3.3V if you supply 3.3V, and are 5V if you supply 5V. Inputs are not 5V tolerant if you supply 3.3V.
- UART: (TX)-Transmit data from ODrive, (RX)-Recieve data to ODrive.
- The isolated inputs don't have a built-in pull-up/pull-down resistor (neither user configurable nor fixed). Setting the GPIO mode to `DIGITAL_PULL_UP` or `DIGITAL_PULL_DOWN` on these pins has the same effect as `DIGITAL`.